

Hard Coral Cover Model

Equation 1

$$C_t^i = \min\left(K^i, \max(0, (C_{t-1}^i + G_t^i + S_t^i)(1 - m^i))\right)$$

with the growth and recruitment terms

$$G_t^i = \max(0, r^i(1 - g^i)) C_{t-1}^i \left(1 - \frac{C_{t-1}^i}{K^i}\right),$$

$$S_t^i = \frac{100 \mu c}{A^i} \left(\sum_j p_{j \rightarrow i} \Lambda_{t-1}^j\right) \left(1 - \frac{C_{t-1}^i}{K^i}\right),$$

the larval supply

$$\Lambda_{t-1}^j = L C_{t-1}^j + R \cdot \mathbf{1}\{t = 1, j \in \mathcal{T}\},$$

and the mortality fraction (capped in code)

$$m^i = \min\left(0.95, P^{i, \text{Typh}} \text{mort}^{i, \text{Typh}} + P^{i, \text{Ble}} \text{mort}^{i, \text{Ble}}\right).$$

Term Definitions

Symbol	Code variable	Definition
C_t^i	<code>current_hcc</code>	Hard coral cover of reef i at step t
C_{t-1}^i	<code>current_hcc</code>	HCC of reef i in the previous step
C_{t-1}^j	—	HCC of source reef j in the previous step
G_t^i	<code>growth</code>	Logistic growth term for reef i at step t
S_t^i	<code>recruit</code>	Recruitment (settled larvae) term for reef i
r^i	<code>R.GROW</code>	Intrinsic growth rate (by reef type)
g^i	<code>PENALTY/p_vec</code>	Gravity penalty on growth (code only), $0 \leq g^i \leq 1$
K^i, K^τ	<code>K_VALS</code>	Carrying capacity / max HCC for reef type τ
m^i	<code>EXP_MORT</code>	Expected mortality fraction (rate on standing cover)
$M^{i, \text{Typh}}$	—	Expected HCC lost to typhoons = $P^{\text{Typh}} \text{mort}^{\text{Typh}} C_{t-1}^i$
$M^{i, \text{Ble}}$	—	Expected HCC lost to bleaching = $P^{\text{Ble}} \text{mort}^{\text{Ble}} C_{t-1}^i$
$P^{i, \text{Typh}}$	<code>typhprob</code>	Typhoon probability at reef i
$P^{i, \text{Ble}}$	<code>bleachprob</code>	Bleaching probability at reef i
$\text{mort}^{i, \text{Typh}}$	<code>typhmort</code>	Per-event typhoon mortality fraction
$\text{mort}^{i, \text{Ble}}$	<code>bleachmort</code>	Per-event bleaching mortality fraction
$p_{j \rightarrow i}$	<code>P.T</code> entry	Settlement probability, source j to sink i
$L^{j \rightarrow i}$	—	Larvae received by i from j (photo term)
Λ_{t-1}^j	<code>larvae</code>	Larval supply from source j (code term)
A^i	<code>AREA</code>	Area of reef i (km ²)
L	<code>LARVAE_PER_HCC</code>	Larvae per unit HCC = 10^6
μ	<code>MATURITY_RATE</code>	Maturity/settlement coefficient = 0.09
c	<code>COLONY_TO_KMSQ</code>	Colony-to-area conversion = 2×10^{-10}
R	<code>RESTORATION_LARVAE</code>	Restoration pulse = 1.4×10^9 (at $t = 1$, targets)
$\mathcal{T}$	<code>target_indices</code>	Set of targeted reefs
$\mathbf{1}\{\cdot\}$	—	Indicator: 1 if condition true, else 0